

A scenic mountain landscape with green hills, rocky cliffs, and a cloudy sky. The foreground shows a rocky ridge with green grass and small yellow flowers. In the background, there are rolling green hills and a deep valley with a river or stream. The sky is filled with large, white, fluffy clouds.

AI & Software Engineering Workshop

Week 1, Day 2: Personality and Prompts

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What is a system prompt?

- An instruction the model sees **before every conversation**
- The user never sees it, but every reply obeys it
- It defines: who the bot is, how it talks, what it refuses to do
- **The most powerful single line in this project**

Same model + different system prompt = a completely different bot.

Find it in the code

bot/config.py:

```
SYSTEM_PROMPT = (  
    "You are a knowledgeable and concise AI assistant. "  
    "Answer clearly and directly. Avoid unnecessary filler. "  
    "Keep responses appropriately brief for a chat interface."  
)
```

Live experiment: change it to *"You answer only in rhymes."*

Restart. Ask it anything.

Build your persona

Pick one, or invent your own:

- 🍳 **Chef**: answers everything with a recipe angle
- 📖 **Tutor**: explains step by step, asks questions back
- 🎲 **Game master**: turns the chat into an adventure
- 🎤 **Comedian**: can't help making it funny

Write 3–5 sentences: who the bot is, how it speaks, one thing it always does, one thing it never does.

Swap the brain

The model is an environment variable in `.env`:

```
AI_MODEL=gpt-oss-120b      # default  
AI_MODEL=zai-glm-4.7       # try this one too
```

- Same persona, different model: what changes?
- Speed vs depth: time the answers, compare the styles

Pair demo

1. Swap seats (or share bot usernames)
2. Chat with your partner's bot for 3 minutes
3. Guess their system prompt from the replies alone
4. Feedback: one thing that works, one thing to sharpen

The faster the other person can guess your prompt, the better it is.

Make the bot match its identity

Two commands still describe the *old* bot:

- `/help` : should list what **your** bot does, in its voice
- `/about` : should introduce **your** persona

Both live in `bot/handlers.py` . Edit them like you edited `/start` yesterday.

Today → Tomorrow

Today your bot has a personality, defined in one string.

Tomorrow: new powers. You'll write your own slash commands, including one that calls the AI itself.